ZYROO Data Analytics Internship

Week 6: Final Advanced BI Product & Project Completion

Project: Ride Analytics & Revenue Intelligence Platform

Project Progress

Week 2: Ride demand, peak hours, locations, weekday/weekend patterns, and customer behavior were analyzed.

Week 3: Revenue patterns and driver performance were analyzed using operational and financial KPIs.

Week 4: The analysis was converted into an interactive Power BI dashboard.

Week 5: The dashboard was upgraded with advanced DAX, segmentation, operational analysis, scenario analysis, and deeper business intelligence.

Final Stage: Week 6 completes the entire project as a polished, validated, optimized, presentation-ready Business Intelligence product. After this week, this project is considered complete and the internship moves to the next project.

Goal: Deliver a fully optimized end-to-end analytics product that can be demonstrated to a business stakeholder and reviewed by another analyst.

1. Final Project Audit

- Review every component created from Week 2 through Week 5.
- Identify and fix broken calculations, duplicated logic, unused fields, inconsistent naming, weak visuals, and incompletedocumentation.
- Create a final issue list and resolve all critical issues before submission.
- Confirm that the final dashboard tells one consistent business story.

2. Final Data Quality & Validation

- Run a final audit for missing values, duplicates, invalid dates, impossible distances, negative or zero fares, invalid ratings,inconsistent statuses, and duplicate IDs.
- Document cleaning and exclusion rules.
- Reconcile key totals with source data.
- Independently validate critical KPIs using SQL, Python/Pandas, Excel, or another suitable method.
- Record remaining limitations.

3. Final Data Model Optimization

- Review the Power BI model for unnecessary columns and tables.
- Use appropriate relationships and correct cardinality.
- Centralize measures and remove duplicate calculations.
- Use a clean Date table for time intelligence where applicable.
- Check filter direction and avoid unnecessary bidirectional relationships.
- Improve naming conventions.

4. Final DAX & KPI Optimization

- Review all major DAX measures for correctness and consistency.
- Remove duplicate or unused measures.

- Optimize expensive calculations where possible.
- Validate time-intelligence calculations and filter behavior.
- Finalize executive KPIs such as total rides, completed rides, cancellation rate, total revenue, average fare, average rating,and revenue per completed ride.
- Document KPI definitions.

5. Executive Dashboard Finalization

- Create a clear executive overview page.
- Use consistent titles, units, number formats, labels, and terminology.
- Reduce visual clutter and remove visuals that do not add decision value.
- Make important findings visible with minimal interaction.
- Ensure slicers and cross-filtering behave correctly.

6. Advanced Analytical Pages

- Finalize Demand, Revenue, Customers, Drivers, and Operations pages.
- Include the advanced analysis developed in Week 5.
- Add drill-through and report tooltips where useful.
- Use conditional formatting only when it improves interpretation.
- Give every page a clear analytical purpose.

7. Final Customer Intelligence

- Finalize customer segmentation and behavior analysis.
- Compare customer frequency, repeat behavior, ride contribution, and revenue contribution where available.
- Identify high-value or high-frequency segments using transparent rules.
- Document limitations if full customer lifetime value cannot be measured.

8. Final Driver & Operations Intelligence

- Finalize driver performance using relevant KPIs.
- Compare driver results with appropriate benchmarks.
- Finalize cancellation and operational analysis by time, location, ride type, and other supported dimensions.
- Highlight areas requiring attention without unsupported causal claims.

9. Final Scenario & Decision Analysis

- Finalize at least two practical scenario analyses where supported.
- Examples include changes in completion rate, cancellation rate, average fare, or completed ride volume.
- Show baseline, scenario assumption, estimated result, and difference.
- Clearly label scenarios as estimates and document assumptions.

10. Business Impact Analysis

- Connect findings to demand, revenue, completion, cancellations, customers, and drivers.
- Prioritize recommendations by evidence and potential business relevance.
- Separate directly supported findings from hypotheses requiring additional data.

11. Final Business Recommendations

- Produce at least 7 recommendations.
- Each recommendation must reference its supporting finding.
- Cover multiple business areas.
- State what should be improved, monitored, investigated, or tested next.
- Identify recommendations requiring additional data before implementation.

12. Final Report

- Create the final Business Intelligence report.
- Include Business Problem, Dataset, Data Preparation, Data Quality, Data Model, Methodology, KPIs, Analysis,Dashboard, Advanced Analysis, Findings, Recommendations, Limitations, and Future Improvements.
- Include important dashboard screenshots.
- Explain major analytical logic in simple technical language.

13. Final Presentation

- Prepare a professional 10 to 15 minute presentation.
- Present problem, data, methodology, analysis, dashboard, findings, recommendations, limitations, and final product.
- Demonstrate the interactive dashboard live.
- Show at least one advanced analysis and one scenario analysis.
- Be ready to explain major KPI calculations and validation.

14. GitHub Production Readiness

- Clean the repository before final submission.
- Remove temporary files, duplicates, unused datasets, and unnecessary outputs.
- Use clear file and folder names.
- Update README with overview, features, tools, workflow, screenshots, findings, and recommendations.
- Add final report and presentation materials.

15. Final GitHub Structure

- ride-analytics-revenue-intelligence/
- data/
- sql/
- python/
- powerbi/
- reports/
- presentation/
- screenshots/
- README.md
- requirements.txt
- .gitignore

16. Final Testing

• Test all dashboard pages, slicers, filters, drill-through, tooltips, and KPI values.

• Test the report in the intended Power BI environment.

• Check spelling, formatting, labels, units, dates, currency, and percentages.

• Confirm no important visual is blank or incorrectly filtered.

17. Final Deliverables

• Fully optimized Power BI dashboard.

• Validated analytical data model.

• Final DAX/KPI layer.

• Advanced demand, revenue, customer, driver, and operations analysis.

• At least 2 documented scenario analyses.

• At least 7 final business recommendations.

• Final Business Intelligence report.

• Final presentation.

• Complete GitHub repository.

• Project screenshots and supporting evidence.

18. Final Submission Checklist

■ Data quality audit completed

■ KPI validation completed

■ Data model optimized

■ DAX measures reviewed and optimized

■ Executive dashboard finalized

■ Demand page finalized

■ Revenue page finalized

■ Customer page finalized

■ Driver page finalized

■ Operations/cancellation page finalized

■ Scenario analysis completed

■ Final insights documented

■ 7+ recommendations documented

■ Final report completed

■ Presentation completed

■ GitHub repository cleaned and updated

■ README completed

■ Final dashboard testing completed

19. Final Success Criteria

• The project works as one complete analytical product rather than separate weekly tasks.

• Important KPIs are validated and consistently calculated.

• The dashboard is interactive, readable, and optimized for decision-making.

• Advanced analysis is connected to real business questions.

• Recommendations are supported by project evidence.

• Documentation allows another analyst to understand and reproduce the workflow.

• The final presentation clearly explains the journey from raw data to business decision support.

20. Project Completion

• Week 6 is the final week for the Ride Analytics & Revenue Intelligence Platform.

• After all final deliverables are submitted and reviewed, this project is considered complete.

• No additional development phase is planned for this project after Week 6.

• The internship will then move to the next project for the final two weeks.

21. Final Project Flow

• Raw Data → Cleaning → EDA → Demand & Customer Analysis → Revenue & Driver Analysis → Power BI Dashboard → Advanced BI → Scenario Analysis → Validation → Optimization → Executive Reporting → Final Presentation → Completed Analytics Product